

PROJECT DESCRIPTION DOCUMENT

Public Sentiment & Opinion Mining System:
Uttarakhand Government Focus

Prepared For: Project Mentor & Stakeholders

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Version: 1.0 (Prototype Specification)

1. Executive Summary

This document outlines the scope, objectives, and architectural design for the development of a Public Sentiment & Opinion Mining System targeting the Uttarakhand State Government. In light of recent Gen Z-led protests, youth mobilizations regarding recruitment irregularities, and general governance feedback, this prototype aims to systematically capture, process, and analyze public sentiment from digital channels (social media, news portals). By leveraging Natural Language Processing (NLP), the system will provide real-time, data-driven insights into the populace's approval, frustrations, and demands.

Core Mission: To replace anecdotal feedback with quantifiable, continuous public sentiment metrics, enabling data-informed responses to youth and citizen grievances in Uttarakhand.

2. Problem Statement & Context

The Uttarakhand government is currently facing heightened scrutiny from its youth demographic. Recent months have seen large-scale protests, particularly in Dehradun, driven by issues such as:

- Alleged recruitment examination paper leaks and backlogs in state commissions (UKSSSC, UKPSC).
- Demands for fair and transparent investigations (e.g., CBI inquiries).
- Concerns over local employment policies, environmental regulations, and migration.

Currently, the government and analysts lack a centralized, automated mechanism to gauge the scale and intensity of these sentiments across digital platforms. Manual monitoring is slow, biased, and incapable of capturing the volume of data generated by the Gen Z demographic.

3. Project Objectives

3.1. Primary Objectives

- **Automated Data Aggregation:** Extract public comments, news snippets, and social media posts relevant to the Uttarakhand Government and specific political figures or policies.
- **Sentiment Classification:** Accurately classify text data into Positive, Negative, or Neutral polarities using state-of-the-art NLP techniques.
- **Trend Analysis:** Monitor spikes in negative or positive sentiment over time, correlating them with specific events (e.g., policy announcements, protest dates).

3.2. Secondary Objectives

- **Aspect-Based Sentiment:** Identify the specific "aspects" driving the sentiment (e.g., distinguishing between anger over "infrastructure" vs. "jobs").
- **Multilingual Handling:** Process and translate regional dialects (Hinglish, basic Garhwali/Kumaoni terms in roman script) to ensure accurate analysis.
- **Stakeholder Dashboarding:** Deliver a clean, interactive prototype UI that visualizes the metrics for non-technical observers.

4. Proposed System Architecture

The prototype will be built across four distinct phases, ensuring modularity and scalability.

Phase 1: Data Extraction (Ingestion Layer)

Targeting primary digital town squares where youth express opinions. Sources include:

- **X (formerly Twitter):** Using keyword tracking (e.g., #Uttarakhand, #UKSSSC, CM's handles).
- **Local News Portals:** Scraping comments and article context from regional digital news outlets.
- **YouTube Comments:** Extracting top comments from regional news channels covering local ground reports.

Phase 2: Preprocessing & Normalization

Raw internet text is unstructured and noisy. This phase involves:

- Removing URLs, special characters, and bots/spam.
- Translating Hindi/Hinglish comments to English via API for unified NLP processing.
- Tokenization and stop-word removal.

Phase 3: NLP & Sentiment Engine (The Brain)

Applying machine learning to derive meaning:

- **Model Selection:** Utilizing pre-trained HuggingFace models (e.g., RoBERTa for sentiment analysis) or VADER for rule-based quick analysis.
- **Entity Recognition (NER):** Identifying key figures (ministers, departments) and entities (locations like Dehradun, Haldwani).
- **Scoring:** Assigning a polarity score (-1.0 to +1.0) to each comment.

Phase 4: Analytics Dashboard

Transforming raw scores into actionable intelligence via an interactive dashboard featuring:

- Overall Government Approval Score (Gauge Chart).
- Sentiment Timeline (Line Graph) mapping emotion against dates.
- Word Clouds of most frequent grievances (e.g., "Paper Leak", "Unemployment").

5. Technology Stack

Component	Recommended Technologies
Data Scraping	Python (BeautifulSoup, Selenium, Apify, Tweepy)
Data Processing/Storage	Pandas, NLTK, SQLite/PostgreSQL (for prototype)
NLP Models	HuggingFace Transformers (RoBERTa), VADER, TextBlob
Dashboard/UI	Streamlit or Dash, Plotly (for interactive charts)

6. Project Deliverables & Milestones

Milestone	Description	Estimated Timeframe
1. Requirement & Architecture Finalization	Approve data sources, define target keywords, finalize tech stack.	Week 1
2. Scraper Development	Build and test data extraction scripts for social media and News platforms.	Week 2 - 3
3. Data Cleaning & Model Integration	Implement text normalization and integrate the sentiment scoring model.	Week 4 - 5
4. Dashboard Creation	Develop the Streamlit UI and connect it to the sentiment database.	Week 6
5. Prototype Review & Tuning	Present to mentor, tune accuracy, handle edge cases.	Week 7

7. Expected Impact

By delivering this prototype, we provide the Uttarakhand government—and analysts—with a tangible tool to shift from reactive governance to proactive engagement. Understanding the precise nuances of Gen Z protests and citizen demands allows for better policy communication, timely interventions, and a measurable framework for gauging public satisfaction over time.